

ELIF DIKMEN

Philadelphia/PA | eelifddikmen@gmail.com | +1 (551) 201-4579 | US Permanent Resident | GitHub | LinkedIn

EDUCATION

Georgia Institute of Technology

Master of Science in Analytics

Yeditepe University

Bachelor of Science in Computer Engineering

Atlanta, GA

Aug 2026-Present

Istanbul, TR

October 2025

TECHNICAL SKILLS

Databases: BigQuery, MySQL, PostgreSQL

Programming Languages: Python, JavaScript, PHP, HTML/CSS, SQL, Dart, Swift

Frameworks/Libraries: Next.js, React, Node.js/Express

Data Visualization: Matplotlib, Plotly, Tableau, Power BI

Machine Learning & Deployment: Keras, TensorFlow, Pandas, NumPy, Seaborn, scikit-learn, PyTorch, FastAPI

EXPERIENCE

Pareto Technologies

Quantitative Research Analyst Intern

New York, NY

Nov 2025 - July 2026

- Developed a **1-Minute Harmonic Rotation** model using Python (Pandas, NumPy) and statistical dispersion analysis to measure price oscillations around the **Volume Point of Control (VPOC)**, identifying that "Normal" regimes exhibit an **average of 29.22 crosses with 20.08 point rotations** to optimize mean-reversion entries.
- Engineered a **Market Regime Classification algorithm** based on **Initial Balance (IB)** width and standard deviation metrics, quantifying that Expanded IB days increase rotation magnitude by 44% and proving the necessity for dynamic, regime-specific stop-loss expansion.
- Designed a **Session-to-Session Directional Probability framework** utilizing conditional probability and Pearson correlation matrices to evaluate lead-lag relationships, establishing a predictive model for how London session momentum statistically influences New York directional bias.
- Executed empirical hypothesis testing on large-scale tick data to challenge volatility assumptions, disproving the theory that high-volatility days reduce frequency by confirming a persistent **31.12 average cross rate** during aggressive price discovery phases.
- Built automated data visualizations and dashboards using **Matplotlib and Seaborn** to map rotational characteristics, transforming raw tick data into actionable risk-management protocols and regime-specific trading adjustments.

Bologna University

AI/ML Intern

Cesena, Italy

Jul 2025 - Oct 2025

- Engineered a high-scalability data acquisition and ingestion pipeline using asynchronous Python to systematically aggregate and process natural-language specifications from **119,274 GPT apps** and **4,592 Actions**, establishing the largest empirical dataset for LLM app privacy research to date.
- Architected a multimodal **RAG framework integrating FAISS (IndexFlatIP) and OpenAI's text-embedding-3-small** for high-performance vector search, using GPT-4o Vision to autonomously interpret and synthesize insights from dynamically generated hierarchical treemaps.
- Developed an automated privacy policy audit engine that cross-references actual data collection manifests against privacy disclosures using SBERT-based semantic matching; identified that third-party Actions collect 6.03% more data on average than first-party counterparts.
- Implemented data normalization and heuristic filtering modules to categorize unstructured JSON metadata into a structured taxonomy of 24 categories and 145 discrete data types, achieving 100% taxonomic integrity and detecting prohibited sensitive data collection in violation of platform policies.
- Optimized backend system performance by designing a FastAPI-based service layer with SHA-256 hash-based caching for vector indices, reducing redundant embedding computations and improving response latency for large-scale dataset queries.
- Quantified ecosystem-wide privacy risks by analyzing data collection across 119k+ applications, providing empirical evidence of inconsistent policy enforcement and surfacing high-risk data patterns through automated statistical analysis and Plotly-based intelligence dashboards.

PROJECTS

Football Match Prediction

Engineering Final Thesis, Yeditepe University

- Built an ML-based match outcome prediction system (Home Win / Draw / Away Win) across 8 major competitions on 2014-2025 historical data; engineered Elo rating deltas, rolling 5-match form, expected-goals (xG) differentials, and odds-implied probabilities as core features from API-Football, Kaggle, and custom BeautifulSoup/requests scrapers.
- Trained and benchmarked **Logistic Regression, Random Forest, and XGBoost** per league with **GridSearchCV, stratified 5-fold cross-validation**, and a strict temporal train/test split (pre-2023 train, 2023+ test) to prevent lookahead leakage; selected the per-league model by macro-F1 rather than raw accuracy to correct for class imbalance in the draw outcome.
- Validated the trained system against 153 real, previously-unseen matches played after the training cutoff, achieving **57.5% accuracy / 54.3% macro-F1** in production-style testing — closely tracking historical test performance and confirming the model generalizes beyond static historical data.
- Extended the system with a real-time in-match model that re-scores predictions every 60 seconds from live score, half-time state, and live odds, reaching **67.8% accuracy / 65.7% macro-F1** in live-match evaluation; cached fixtures and logged predictions to SQLite to cut redundant API calls and latency.
- Built a native macOS/iOS client in Swift/Xcode surfacing win/draw/loss probabilities, feature-importance explainability views, and live standings; owned the system end-to-end from data ingestion through the model-serving API to the mobile front end.

Exam Planning System

- Designed and implemented a data-driven academic management system to streamline exam scheduling, course assignments, and workload distribution across departments.
- Developed relational database schemas and applied multi-stage normalization to ensure structural integrity, reduce redundancy, and maintain consistent data records.
- Utilized advanced SQL analytics, joins, aggregations, and conditional logic to compute workload metrics, analyze staff utilization, and support automated scheduling decisions.
- Built PHP-based interfaces that visualize SQL outputs and provide actionable insights for academic managers, including workload balancing through real-time assistant assignment scoring.
- Integrated stored queries, procedural SQL logic, and automated reporting workflows to support faculty performance tracking and data-driven administrative decision-making.